

Shubham Saluja Kumar Agarwal

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EDUCATION

Cornell University | Ithaca, NY

August 2026 – May 2027

Master of Engineering in Electrical and Computer Engineering

Purdue University | West Lafayette, IN

September 2022 – May 2026

Bachelor of Science in Computer Engineering

Concentrations: Computer Systems; Microelectronics and Semiconductors

Minors: Mathematics; Business Economics

GPA: 4.00/4.00

RESEARCH EXPERIENCE

Versatile Electronic Systems Lab | Cornell University

September 2026 – Present

Supervisor: Professor Amal El-Ghazaly

Magnetically Actuated Haptic Displays

- Designing new form factors of a soft actuator that uses electro-permanent magnets for actuation.

Hybrid Body Lab | Cornell University

August 2026 – Present

Supervisor: Professor Cindy Hsin-Liu Kao

EdemaFlex – Hand edema treatment glove

- Manufacturing knit gloves with shape memory alloy springs for hand-edema treatment.
- Updating user interface for optimized design file generation.

Responsive Environments Group | Massachusetts Institute of Technology

May 2026 – Present

Supervisor: Professor Joseph A. Paradiso

FiberCircuits – System-on-Fiber, Sleep Assistance Technology

- Creating the next version of the FiberCircuits topology, reducing the width of the fiber from 1.4mm to under 1mm, with components and topologies for applications like sleep detection or electrostimulation.

Nanomaterials Lab | Universidad Autónoma del Estado de Morelos

May 2026 – August 2026

Supervisor: Prof. Vivechana Agarwal

Machine Learning applied to limited materials science datasets

- Trained a time-agnostic machine learning model for small-sample materials datasets to classify contaminants with an accuracy of 97% and predict concentration with a mean average error of 0.061.

QuMETA NanoML Group | Purdue University

March 2025 – Present

Supervisor: Prof. Alexander V. Kildishev, Prof. Alexandra Boltasseva, Prof. Vladimir Shalaev

Inverse Design of Photonic Devices

- Curated and validated a novel metagrating dataset of 10,000+ device structures, with the goal of open sourcing the data to accelerate community research in inverse design.
- Established performance benchmarks for a novel inverse-design ML framework by training 4 distinct baseline models on two custom datasets. Manuscript in progress.
- Implemented a VGGNet-based neural network to achieve high-fidelity predictions and eliminating the need for electromagnetic solvers, reducing efficiency evaluation time by over 100 times with minimal loss in accuracy.

Wireless Sensing Circuits and Systems Laboratory | Purdue University

April 2023 – May 2026

Supervisor: Prof. Byunghoo Jung

AirVolt – Wireless Voltage Sensing

August 2025 – May 2026

- Leading early-stage development of a wireless HVAC voltage-sensing system for use in power plants and power rail installation and maintenance.
- Outlined circuit architecture and designed initial PCB prototypes in Altium Designer for proof-of-concept testing.

Wireless Battery State of Charge Estimation

April 2023 – August 2025

- Developed a method for wireless online LiPo battery charge detection achieving relaxation-induced error under $\pm 1\%$, superior to existing state-of-the-art online sensing methods.
- Implemented an STM32-based frequency counter (30 MHz range, ± 3 Hz error) with a Python GUI for real-time sensor data visualization and logging.
- Created acquisition pipelines for multi-instrument battery datasets, applying filtering, noise removal, and synchronization techniques to extract reliable parameters and validate sensor/device performance.

Vertically Integrated Projects Computer Vision for Embedded Systems | Purdue University

August 2023 – May 2024

Supervisor: Prof. Yung-Hsiang Lu

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Coordinating the 2024 Low Power Computer Vision Challenge (LPCVC)

- Developed a 3D reconstruction evaluation pipeline for the 2024 Low Power Computer Vision Challenge (LPCVC).
- Benchmarked open-source models to define power and accuracy performance baselines for competitor submissions.

PUBLICATIONS AND PRESENTATIONS

- **Kumar Agarwal, S. S.**, Iyer, V., et al. *Benchmarking of Generative Models for Photonic Device Design*. Manuscript in Progress.
- **Kumar Agarwal, S. S.**, Avalos, F. et al. *Time Agnostic Machine Learning Model for Contaminant Classification and Contaminant Concentration Prediction*. Manuscript in Progress.
- Mirani, A., **Kumar Agarwal, S. S.** (presenter), et al. *Transformer for Video Dehazing*. Poster presented at *Purdue 2024 Spring Research Talk & Poster Conference*.
- **Kumar Agarwal, S. S.** (presenter), Vidisha Singhal, et al. *Low-Power Computer Vision Challenge*. Research Talk presented at *Purdue 2023 Fall Research Talk & Poster Conference*.
- **Kumar Agarwal, S. S.**, Mendoza Enríquez, B. U., & Hernández Rivera, D. (2022). *Device Fabrication from Recycled Electronic Spare Parts: Dip Coating Device and High Voltage Power Supply Adapted for Electrospinning Device*. Journal of Physics: Conference Series, 2261, 012001. ICIEE 2022. DOI: 10.1088/1742-6596/2261/1/012001. Presented at the 2022 11th International Conference on Information and Electronics Engineering (ICIEE)

TEACHING EXPERIENCE

Bechtel Innovation and Design Center Peer Mentor

November 2022 – Present

- Directed 300+ student electronics projects from design to manufacturing, guiding diverse teams and people through PCB design, circuit prototyping, debugging, and microcontroller integration.
- Taught and mentored peers in topics ranging from basic soldering and wiring to advanced circuit troubleshooting and embedded programming, adapting lessons to varying skill levels.
- Tackled diverse technical issues by self-learning unfamiliar systems and integrating new knowledge into project solutions.

TECHNICAL AWARDS AND SKILLS

- Awards and Scholarships:
 - Lester B. Knight Scholar for the 2026-2027 Cornell Academic Year
 - Best Electrical and Computer Engineering Senior Design Project at the 2026 Purdue Spark Challenge
 - J. Bruce Majerus Electrical and Computer Engineering Student Scholar Award 2025
 - Honorable Mention for Electrical and Computer Engineering Undergraduate Research Awards 2024
 - Purdue Engineering Student Council Gold Industrial Roundtable Scholarship 2023
 - Best Presenter at 2022 11th International Conference on Information and Electronics Engineering.
- Data and Machine Learning: Python (Numpy, Pandas, PyTorch, Tensorflow), C, MATLAB, SQL
- Design and Simulation Tools: Autodesk Fusion 360, Altium Designer, KiCad, LTSpice
- Digital and Embedded Systems Design: SystemVerilog, RISC-V, Embedded C, Cadence Virtuoso
- Front-End Development: HTML, CSS, JavaScript
- Languages: English, Spanish (native); Hindi (conversational); French (basic)